Christos Hadjinikolis

Engineering Manager | Data Platforms & Production ML
Ph.D. Computer Science | UCL Research Associate | AI Standards
✉ christos.hadjinikolis@gmail.com

profiles
website
LinkedIn
GitHub
Google Scholar

leadership
team development
hiring/coaching
technical direction
delivery ownership

data systems
Kafka/Flink
Spark/PySpark
Airflow
data quality

ML lifecycle
PyTorch/MLflow
time-series models
evaluation/replay
model/data versioning

engineering
Python/Java/Kotlin
SQL/pandas
AWS/GCP
CI/CD/Docker

practice
design reviews
data contracts
local E2E tests
observability

profile

Engineering Manager with 16 years across AI research, data engineering, applied ML and consulting. I lead cross-functional teams building reliable data pipelines and production prediction systems from noisy, real-world telemetry. My experience combines people development, technical direction, model evaluation and MLOps, with earlier team leadership on vehicle predictive maintenance and a Ph.D. in Computer Science.

experience

Vortexa Ltd, London, UK 12/2020–present — Engineering Manager | ML Systems Lead

Lead technical direction and delivery for a live ML/data estate processing **~6M vessel-position records/hour** for **13.5K monitored vessels**. Manage **6 direct reports** and lead a **10-person cross-functional team**. Joined as Senior ML Engineer; Engineering Manager / ML Systems Lead since **02/2022**.

- **People & Team Development:** Own performance reviews, career development and delivery across Data Science, Data Engineering, Product and domain analysts. Hiring manager for **6 roles**; contributed hiring practices, onboarding and engineering standards as Data Production grew from **4 to 30+**.
- **ML Lifecycle:** Led research-to-production delivery for destination and arrival-time sequence/transformer models in **PyTorch**; established MLflow, model/data versioning, automated evaluation gates, monitoring and versioned model-serving workflows.
- **Evaluation & Data Quality:** Built batch/online evaluation and replay loops; analysed edge cases and failure modes with domain experts to guide model and data improvements. Maintained signal-quality controls and live prediction-accuracy monitoring.
- **Pipeline Architecture:** Led the **Kafka Streams-to-Flink** transition to improve maintainability, operational visibility and compute scaling independent of Kafka partitioning; connected streaming enrichment, inference services and downstream data products.
- **Iteration Speed & Reliability:** Standardised I/O, schema/data validations and local end-to-end tests using characteristic sample data; cut a **3-hour DAG feedback loop to under 5 minutes**. Established runbooks, rollback/fallback paths and shared on-call practices.
- **Technical Roadmap:** Ran workshops across Product, analysts and engineers to resolve model-quality disputes, define measurable objectives and prioritise delivery. Used design reviews and architecture decisions to make trade-offs explicit and align teams.

DataReply, London, UK 04/2016–12/2020 — Senior Consultant | Data Scientist / ML Engineer

Joined the London spin-off as its first consultant and contributed to its growth to **30+ consultants**. Progressed to Senior Consultant, leading client delivery, scoping technical roadmaps, mentoring colleagues and coordinating consultants across enterprise engagements.

- **CNHi | Vehicle Telemetry:** Lead Data Scientist/Scrum Master for predictive maintenance on **Azure/Databricks**. Guided a DS/DE team using **PySpark** over live agricultural-vehicle sensor data; developed time-series models for maintenance/repair alerting and translated stakeholder requirements into the delivery backlog.
- **Vodafone Group | ML Platform:** Led multi-workstream delivery on a **GCP/Kubeflow**-based data-science platform, covering migration planning, CI/CD, workflow automation, PySpark feature engineering and ML lifecycle support.
- **UBS | Streaming Analytics:** Delivered real-time **Kafka**/Elasticsearch pipelines, graph/process-mining analytics and observability; developed production ML engineering practice through XP and pair programming.

KCL, UCL, David Game College, UK 2010–2016 — Teaching and Technical Training

Taught algorithms, data structures, databases/SQL and Java/Python during and after doctoral research, building a sustained foundation in technical communication and mentoring.

education

2010–2014 **Ph.D. in Computer Science** King's College London, UK

***Thesis**: Persuasion Dialogues & Opponent Modelling.*

Developed logical inference methods for large knowledge graphs, combining symbolic reasoning, graph analytics and **Bayesian techniques (MCMC)**. Received **Best Poster Award at IJCAI 2013** (413 submissions), EPSRC PhD Scholarship, Outstanding TA Award and Graduate Certificate in Academic Practice.

Publications on Google Scholar

2004–2010 **Diploma (BEng) in Computer Engineering** University of Thessaly, GR

Five-year polytechnic degree with a strong mathematics foundation. Majored in Artificial Intelligence.

2002–2004 **Military Officer: Second Lieutenant** National Guard, CY

Completed national service as a commissioned Second Lieutenant.

research & AI standards

since 10/2024 **University College London, Dept. of Information Studies** Associate Researcher

Research and teach practical AI standardisation, including auditability, model/data versioning, explainability and responsible adoption.

since 2021 **ISO / CEN-CENELEC AI standardisation** Committee Expert | JTC 21, WG3

Contribute to AI standards aligned with EU policy and international norms, with emphasis on traceability, explainability and model/data governance.

selected public engineering tools

PyPI **dynamicio** Data contracts & pipeline testing

Created a Python library that makes I/O boundaries explicit, validates schema/data expectations and switches local/dev/prod datasets. Used across **15+ repositories** to decouple pipeline logic from infrastructure and support faster local feedback.

PyPI **skeleton-replay** Runtime evidence

Python developer tool that captures script/pytest runs as traces, architecture snapshots and replayable reports for debugging, review and onboarding.

JetBrains **Skeleton Replay plugin** Developer workflow

PyCharm/IntelliJ plugin bringing runtime reports, workflow evidence and source navigation into the IDE.

selected talks & interviews

2023 **Agile in Action** Podcast Interview

Agile Data Science: the engineering and delivery journey at Vortexa.

2022 **Open Data Science Conference** Industry Conference Talk

dynamicio: why abstracting I/O is a foundation for MLOps.

2020 **Big Data Warsaw** Industry Conference Talk

Monitoring and analysing communication and trade events as graphs.

selected credentials

2016–2020 **ML, cloud & data engineering** AWS, Google Cloud, O'Reilly

AWS Machine Learning Specialty (2020); Google Professional Data Engineer (2017); Apache Spark Developer (2016).

2016–2019 **Data & graph analytics** Coursera, Elastic, Neo4j

Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.